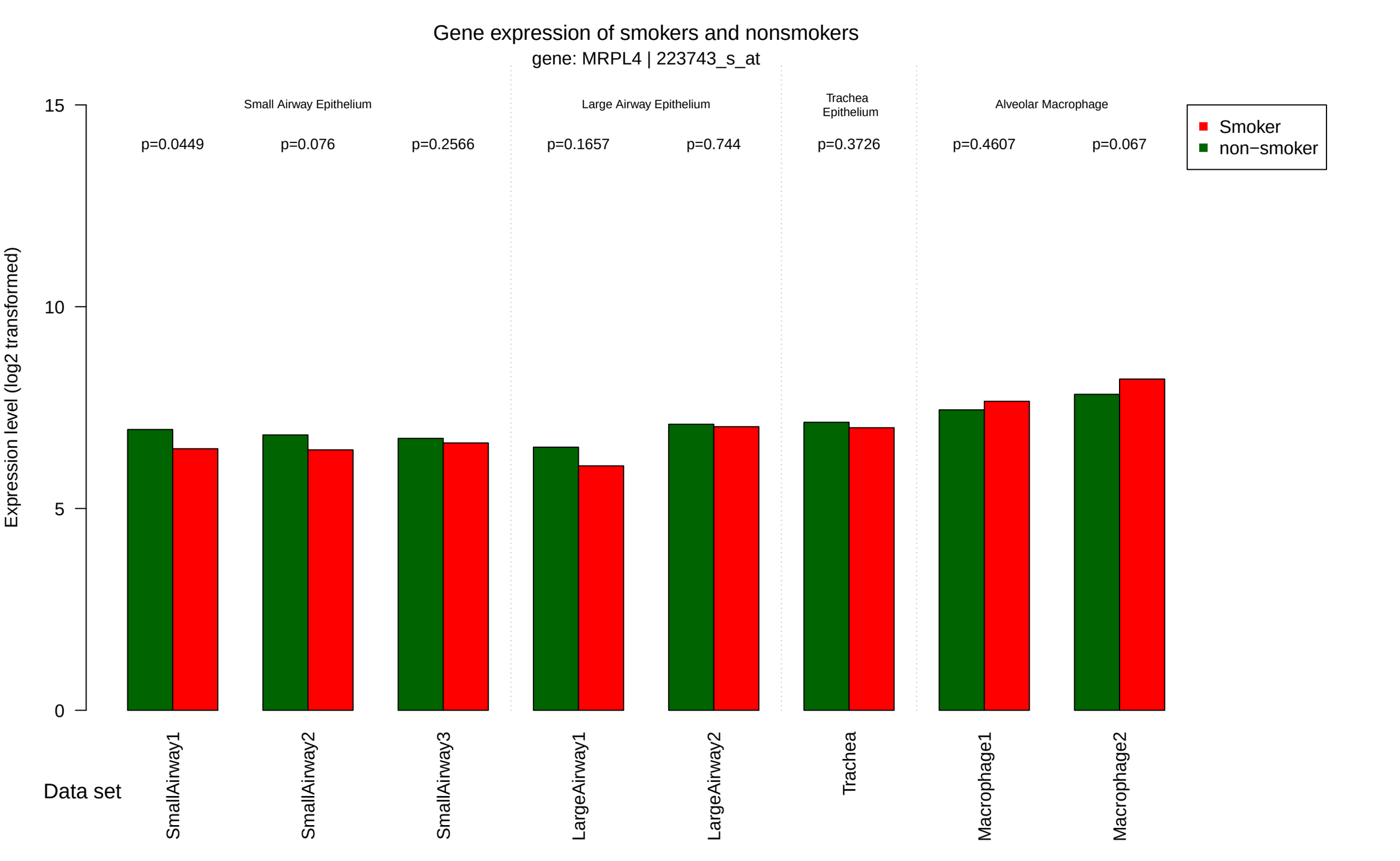
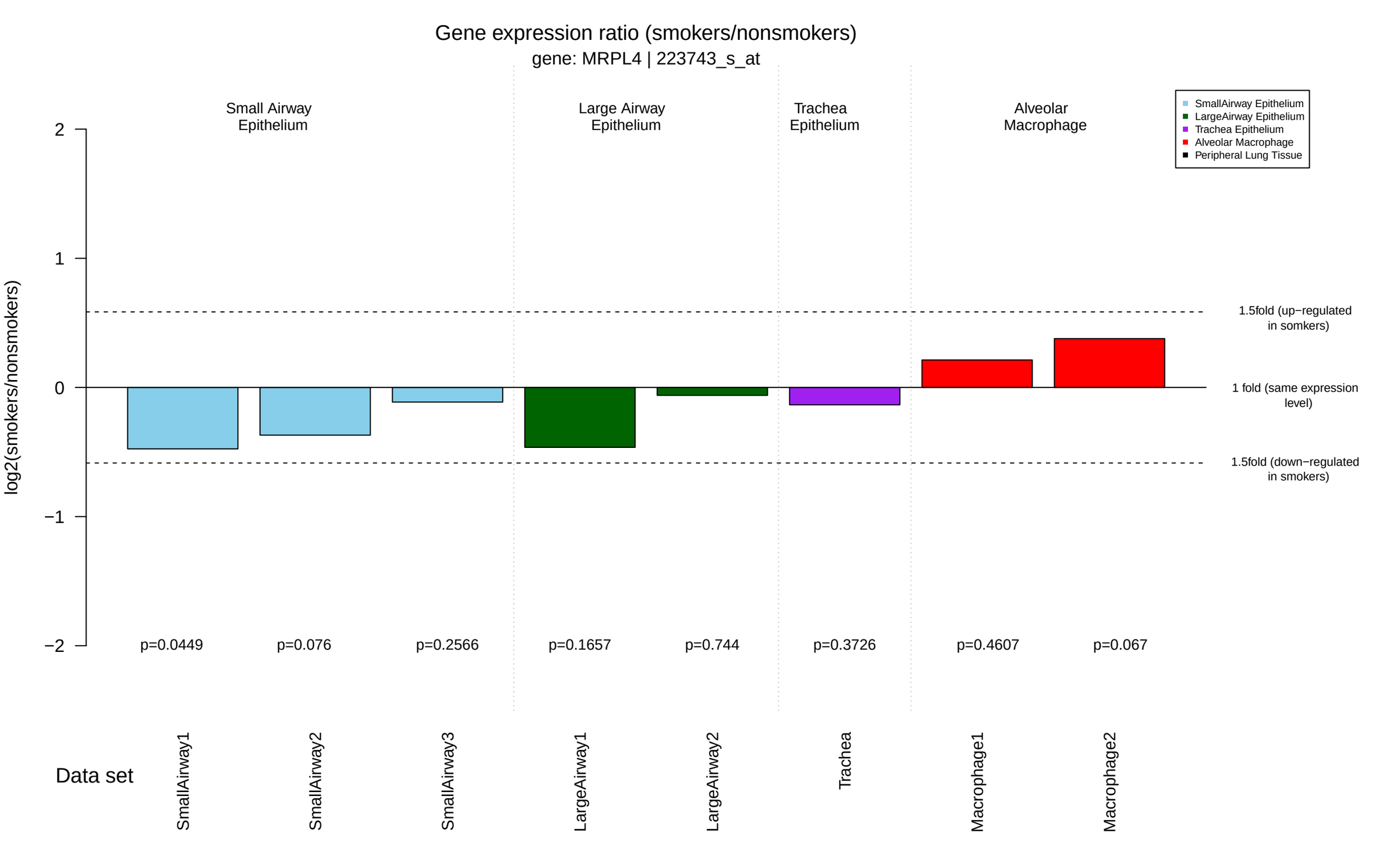
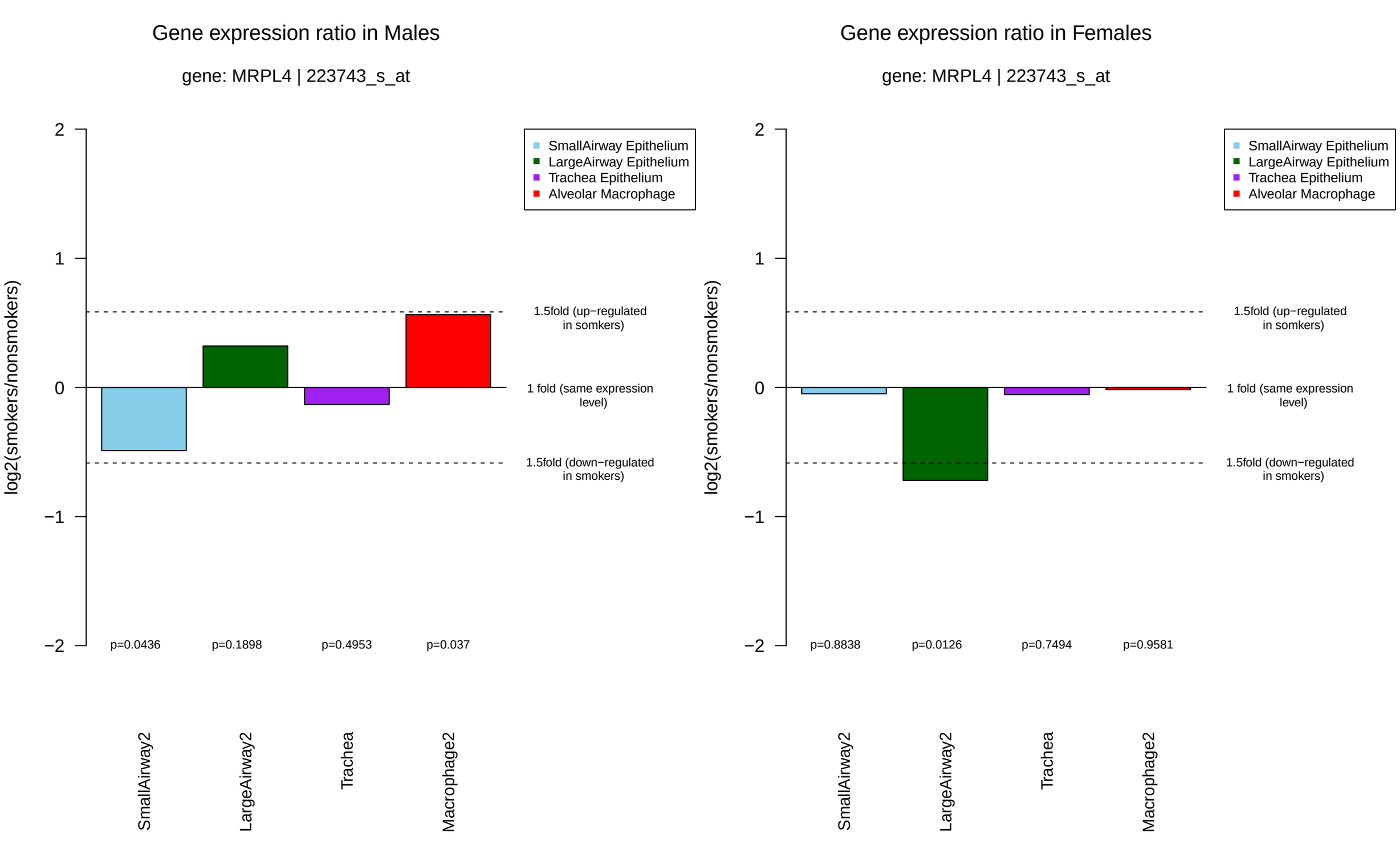
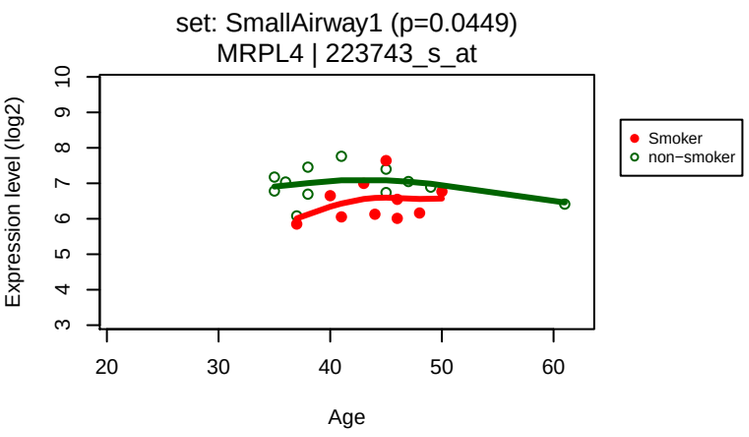
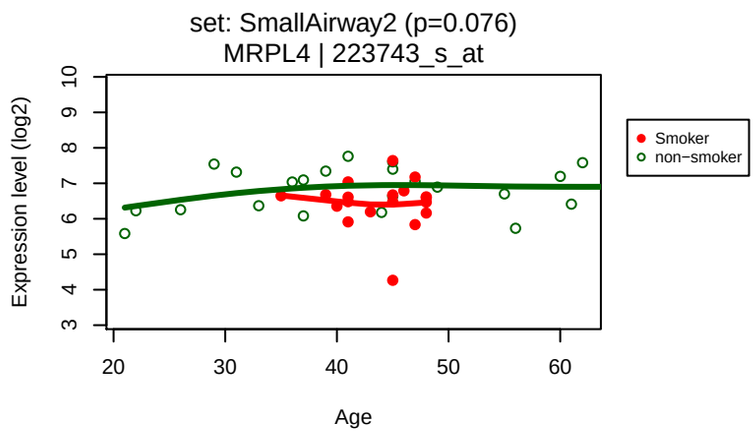
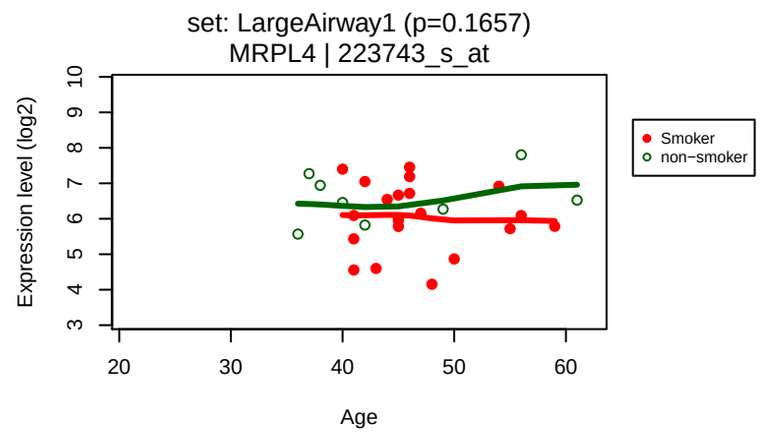

Gene expression of smokers and nonsmokers
gene: MRPL4 | 223743_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0449
p=0.076
p=0.2566
p=0.1657
p=0.744
p=0.3726
p=0.4607
p=0.067
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

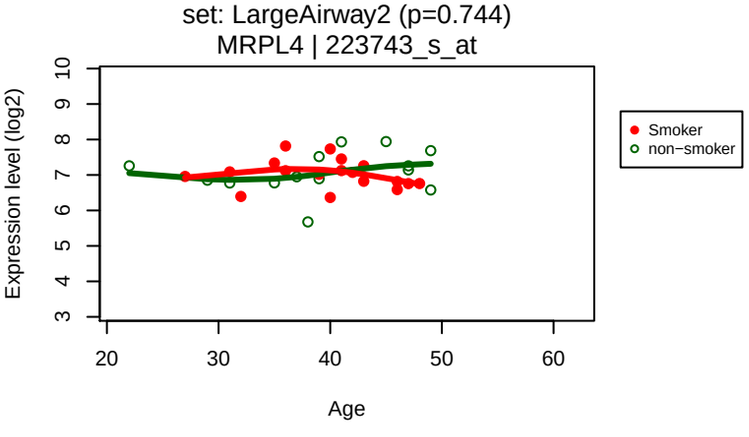

Gene expression ratio (smokers/nonsmokers)
gene: MRPL4 | 223743_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0449
p=0.076
p=0.2566
p=0.1657
p=0.744
p=0.3726
p=0.4607
p=0.067
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

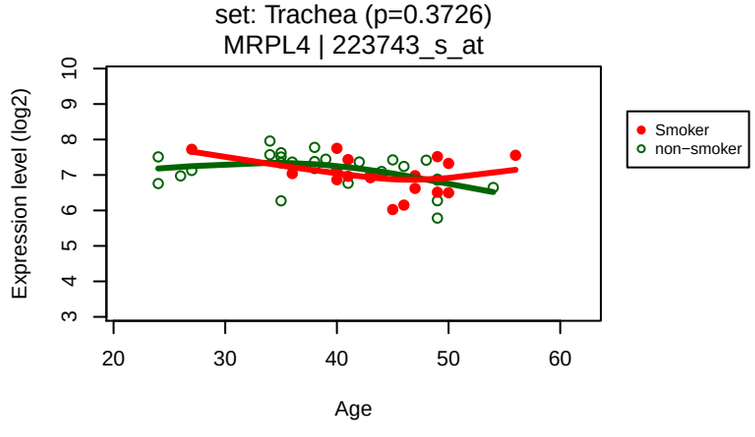

Gene expression ratio in Males
gene: MRPL4 | 223743_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0436
p=0.1898
p=0.4953
p=0.037
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL4 | 223743_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8838
p=0.0126
p=0.7494
p=0.9581
SmallAirway2
LargeAirway2
Trachea
Macrophage2

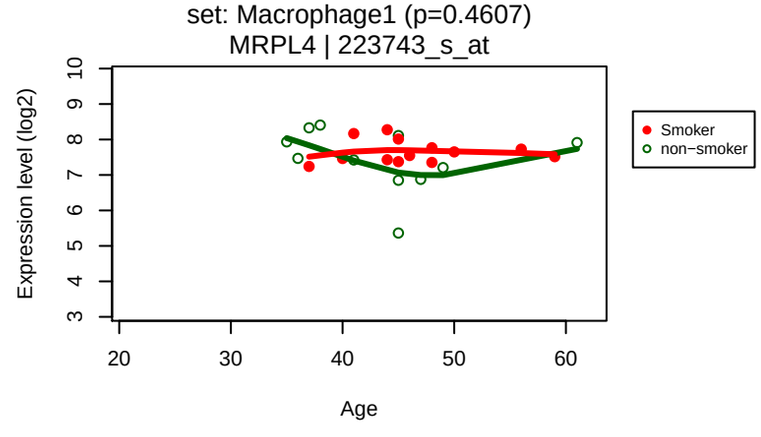

set: SmallAirway1 (p=0.0449)
MRPL4 | 223743_s_at
Expression level (log2)
Age
Smoker
non−smoker

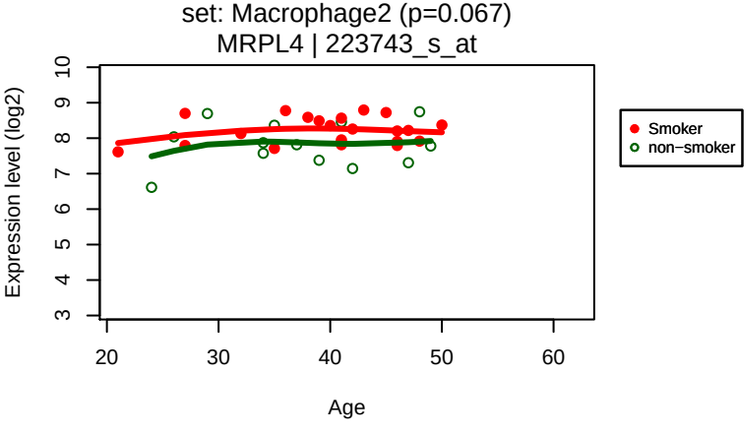

set: SmallAirway2 (p=0.076)
MRPL4 | 223743_s_at
Expression level (log2)
Age
Smoker
non−smoker

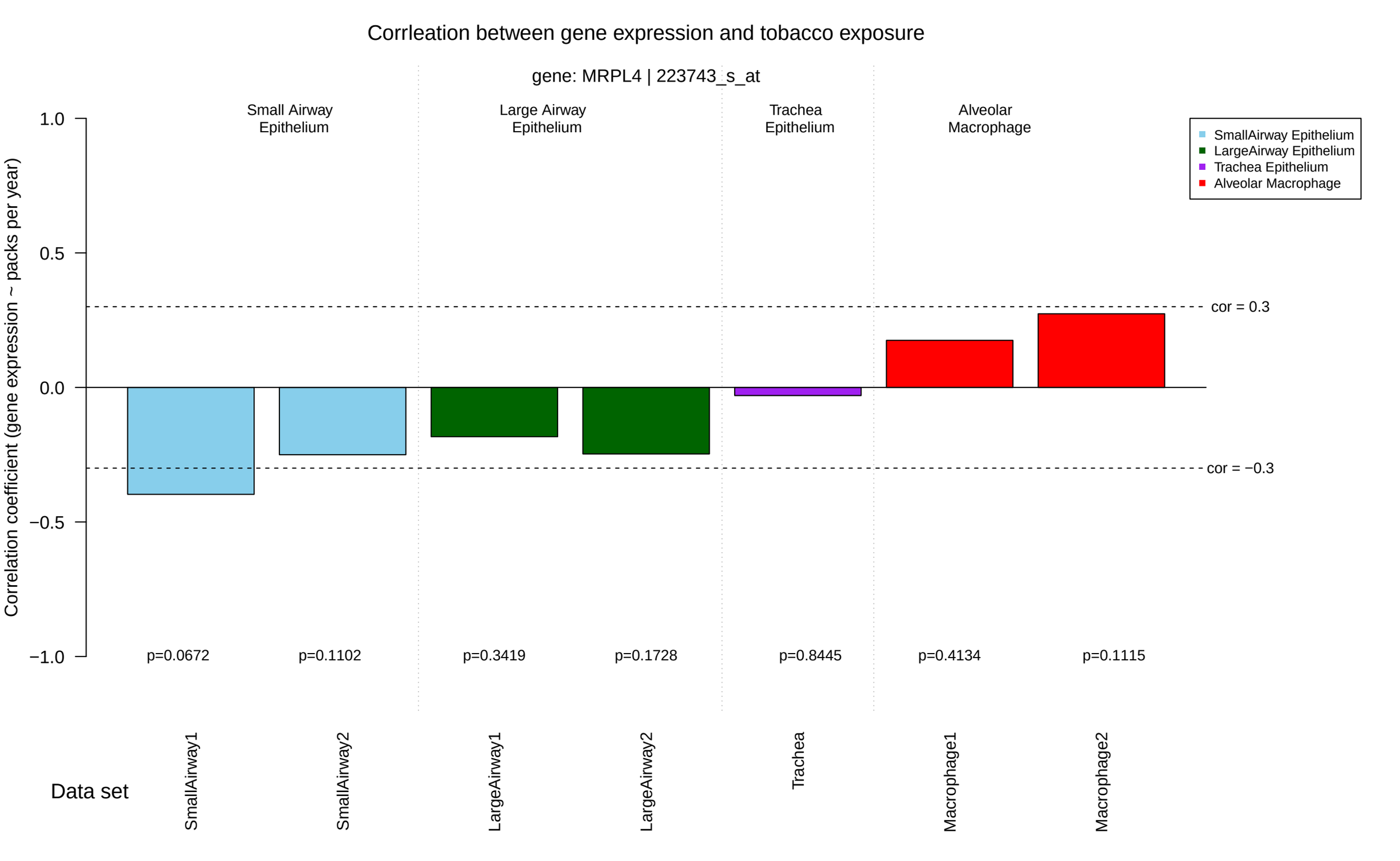

set: LargeAirway1 (p=0.1657)
MRPL4 | 223743_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.744)
MRPL4 | 223743_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3726)
MRPL4 | 223743_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4607)
MRPL4 | 223743_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.067)
MRPL4 | 223743_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL4 | 223743_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0672
p=0.1102
p=0.3419
p=0.1728
p=0.8445
p=0.4134
p=0.1115
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2